



anything else that can be further improved and expanded upon look at studying and learning and work completion tools and also lms tools also want grading tools

Yes—Semester can be substantially strengthened through a full **Learning, Work Completion, Assessment, and Gradebook system**. Since your vision includes a native LMS, the opportunity is not simply to reproduce assignments and grades; it is to build a source-aware, accessible, feedback-rich learning environment that helps students complete work well while giving faculty trustworthy assessment and grading controls.

The market-standard foundations should include LTI 1.3/LTI Advantage for LMS interoperability, Assignment and Grade Services for gradebook exchange, and QTI 3 for portable assessment content. QTI 3 explicitly supports exchange of items, tests, usage data, results, technology-enhanced interactions, and improved accommodations/accessibility. ^[1599]
[1600] ^[1]

Learning and study tools

1. Course Learning Hub

Every course needs a calm, structured workspace—not a file dump.

```
Course Home
├─ What matters now
├─ Course roadmap
├─ Modules and learning outcomes
├─ Assignment timeline
├─ Study plan
├─ Materials and source library
├─ Lecture notes and recordings
├─ Practice and retrieval
├─ Discussions and collaboration
├─ Assessments
├─ Grades and feedback
├─ Office-hours and help routes
├─ Course AI policy
└─ Course accessibility options
```

The student should always see:

Next action
Due date and time zone
Estimated effort, clearly labelled as an estimate
Course importance/weight
Source and instructor owner
Submission state
Feedback state
Recovery/support route

2. Syllabus-to-course workspace

With instructor approval, Semester can turn a syllabus into a reviewable course structure.

Upload syllabus

- extract dates, modules, grade weights, policies, learning outcomes, required materials, office hours, accessibility notes, and AI policy
- instructor reviews each extraction
- publish approved items to the course
- students receive source-anchored timeline and checklist
- changes create visible version history and notification

Do not silently convert an uploaded syllabus into authoritative course rules. The instructor or designated course owner must approve the extracted structure.

3. Work completion planner

This is one of Semester's strongest student features.

Assignment brief

- requirements checklist
- rubric translated into student-friendly criteria
- milestones
- time-block suggestions
- source/material checklist
- draft workspace
- citation/help tools
- peer/instructor feedback route
- final submission checklist
- submission confirmation
- feedback and revision plan

Assignment breakdown

Assignment:
Research paper, 25% of grade

Semester suggests:

- Read prompt and rubric
- Confirm topic
- Locate 5 eligible sources

- Build outline
- Draft introduction
- Draft body sections
- Review citations
- Run accessibility/document-format check
- Compare against rubric
- Submit
- Save proof of submission

Student controls:

- Accept/edit/remove milestones
- Set work times
- Pause reminders
- Mark support needed
- Share selected draft or question

4. Draft and writing workspace

Outline builder
 Focus/distraction-minimized writing mode
 Version history
 Autosave and recovery
 Comments and feedback
 Citation manager integration/export
 Source cards and quotations with page anchors
 Academic-integrity and course-AI-policy panel
 Accessibility checks
 Document-format and submission checklist
 Export to DOCX/PDF/approved LMS format

Do not make AI rewriting the default. Provide clear modes:

Plan
 Explain
 Ask questions
 Check requirements
 Give feedback
 Cite sources
 Improve accessibility
 Draft with disclosure where allowed

5. Study Studio

Study Studio should turn authorized materials into active learning.

Source-grounded summaries
 Vocabulary builder
 Flashcards and spaced-repetition schedule
 Practice questions
 Teach-back mode
 Concept maps

Worked-example mode
Problem decomposition
Formula/reference sheet
Lecture-note organizer
Audio notes and transcript tools
Study-session planner
Focus timer, optional
Confidence reflection
Tutor/library/writing-center handoff

Each generated item should display:

Source:
Lecture 4 slides, pages 12-19

Status:
Generated study aid—review against original material

Course policy:
AI study support permitted for private practice

6. Course-resource intelligence

Help students find the right materials without exposing private content.

Search by concept, week, outcome, assignment, reading, or question.
Show source type: instructor file, library source, student note, official policy.
Show access rights and expiry.
Show “used in this assignment” relationships.
Show citations/page anchors.
Flag duplicate/outdated versions.
Recommend library/tutor/office-hour support when source coverage is weak.

Assessment and grading tools

7. Assessment builder

Faculty need to create assessments without needing separate tools.

Question banks
Reusable item libraries
Question versioning
Tags: course, outcome, topic, difficulty, cognitive level, accessibility review
Shared stimuli
Rubrics
Assessment blueprints
Randomized item pools
Sections and rules
Practice versus graded mode
Question preview
Student preview

Accessibility preview
QTI import/export

QTI is the most important assessment interoperability route because it supports exchange between item authoring systems, item banks, test construction tools, delivery systems, scoring engines, and learning platforms. [¹⁵⁹⁹][1603]

Question types

Multiple choice
Multiple response
True/false
Matching
Ordering
Fill-in-the-blank
Short answer
Essay
Numeric and formula response
File upload
Audio/video response
Coding response, sandboxed
Case/scenario response
Hotspot/image interaction, with accessible equivalent
Table/matrix response
Peer review
Group submission
Portfolio submission
Oral presentation/checklist
Offline/manual assessment record

Every non-text interaction must have an accessible alternative or equivalent pathway.

8. Assessment delivery engine

Timed and untimed delivery
Availability windows and time zones
Multiple attempts
Practice attempts
Question randomization
Section sequencing
Save/resume
Autosave
Submission receipt
Late-policy configuration
Accommodation-aware timing and availability
Make-up/alternative assessment workflow
Offline contingency instructions
Secure-browser/proctoring integration only by institutional decision
Incident/report issue button
Accessibility preferences

Timed online testing can create accessibility barriers, so Semester should support institution-authorized accommodations, flexible assessment design, practice environments, accessible formats, and alternatives rather than assuming a single timed format works for everyone. ^[1606]
[1607]

9. Rubric engine

Rubrics should be first-class structured data, not a PDF attachment.

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Criterion
Performance levels
Points or qualitative scale
Learning-outcome mapping
Instructor annotations
Student-facing language
Exemplars
Accessibility notes
Reusable templates
Version history
Calibration mode
Rubric preview
Student self-assessment
Peer-review version, if enabled
```

10. Gradebook

Build a gradebook that is transparent to students but powerful for instructors.

```
Gradebook capabilities:
- Assignment groups and weights
- Points, percentages, letters, competency/mastery scales
- Drop-lowest rules
- Excused work
- Manual overrides with reason/audit
- Late-policy rules
- Missing/incomplete status
- Grade release controls
- Anonymous grading mode
- Group grading and individual adjustments
- Rubric-linked scoring
- Comment/audio/video feedback
- Import/export
- Grade history
- Final-grade calculation preview
- Student view preview
- SIS/LMS sync controls
```

Student grade view

Current standing:

Estimated, based on released grades only

Included:

Assignments and weights currently released

Not included:

Unreleased work, future assessments, instructor adjustments, and institution-specific final-grade rules

What changes this:

Upcoming assessment weight, missing work, replacement rules, drop-lowest rule, or instructor updates

Next action:

Review feedback on Essay 1 and attend office hours if needed

This protects against a common gradebook failure: presenting a speculative number as an official final grade.

11. Grading workflow

Submission arrives

- automated receipt and integrity/accessibility checks
- optional anonymous grading assignment
- rubric scoring
- feedback comments, audio, video, or annotations
- instructor/TA calibration and moderation
- grade review
- controlled release
- student receives source-linked feedback
- student asks clarification or requests review through policy workflow
- grade-change audit trail
- optional sync to official LMS/SIS

Grading fairness controls

Anonymous grading option

Randomized grading order

Rubric-first scoring

Calibration sets for multiple graders

Inter-rater agreement view

Blind double-marking, where configured

Moderation/second-review workflow

Grade-change reason and audit trail

Student-visible feedback timing

Appeal/regrade request workflow

Accommodation-aware assessment records

No AI-only final grade for consequential work

12. AI in assessment and grading

AI can assist, but it should not silently become the grader.

Use case	Permitted role	Required control
Rubric drafting	Suggest criteria and language	Instructor review before use
Question generation	Draft questions from approved source material	Faculty review; accuracy, bias, and accessibility checks
Practice feedback	Give formative feedback, hints, and explanations	Clearly label as AI; cite course sources; no official grade
Writing feedback	Identify clarity, structure, citation, and rubric gaps	Student review; course-policy compliance; no invisible rewriting
Instructor feedback draft	Draft feedback from instructor rubric inputs	Instructor edits/approves before release
Grading assistance	Flag possible rubric evidence or calculate deterministic scores	Human validates; preserve decision/audit record
Integrity support	Highlight unusual patterns or citation concerns	No automatic misconduct determination; human investigation
Accessibility support	Generate first-pass alt text, transcript aids, plain-language rewrite	Human validation; preserve academic meaning
Final grading	Not AI-only	Instructor/authorized human remains responsible

13. Academic-integrity layer

Build integrity as clarity and learning support, not surveillance.

Course-specific policy panel
Assignment-specific AI disclosure rules
Citation expectations
Version history for student work
Student authorship/process reflection
Permitted tool disclosure
Source/citation checks
Similarity/originality integration, if institution chooses
Instructor review queue
Student explanation and appeal pathway
Integrity education modules
No automatic accusation or penalty

14. Learning analytics—ethical version

Useful faculty and institution insight:

Module completion patterns, aggregated where appropriate
Assessment item quality
Question difficulty and discrimination metrics

Rubric/learning-outcome performance
Feedback turnaround time
Resource and help-route usage
Accessibility issue reports
Course-content freshness
Assessment technical errors

Prohibited or highly restricted:

Hidden student-risk scores
Automated academic intervention based solely on click behavior
Employer access to learning analytics
Emotion/sentiment surveillance
Opaque engagement ranking
Use of accommodation/basic-needs data in performance prediction

LMS interoperability and architecture

Standards and integrations

Need	Recommended approach
Launch into external LMS	LTI 1.3 / LTI Advantage
Course roster/roles	LTI Names and Role Provisioning Service or OneRoster
Gradebook exchange	LTI Assignment and Grade Services
Assessment portability	QTI 3.0
Outcomes/competencies	CASE, when competency workflows are mature
Identity	OIDC/SAML and SCIM
SIS grades	Institution-approved API/integration, preview and audit required
Calendar	iCal/CalDAV or approved Google/Microsoft connector
Content migration	Imports with source/ownership/rights validation
Analytics	Privacy-governed event architecture; Caliper only when justified

LTI Assignment and Grade Services supports creation of gradebook columns and posting of scores associated with a resource link. Semester should treat every external grade write as an explicit, previewable, auditable action—never a silent background sync. ^[2] ^[1]

Critical backend services

Identity and tenant context service
Course/enrollment/role service
Content and versioning service
Assignment and submission service
Assessment/QTI service
Attempt/timing/accommodation service

Rubric and grade-calculation service
Feedback and annotation service
Gradebook and grade-history ledger
Integrity and policy service
AI policy/retrieval/evaluation service
Notification/reminder service
Calendar/task sync service
Accessibility preference and accommodation-application service
Analytics service with privacy controls
Integration gateway for LTI, OneRoster, SIS, and external LMS
Audit/evidence/retention service

Gradebook data model

Course
Enrollment
Assignment group
Assessment/assignment
Gradebook column
Grading scheme
Rubric and rubric version
Submission and submission version
Attempt
Accommodation application
Grade entry
Grade calculation
Feedback item
Grade release event
Override/adjustment
Grade-change request
Appeal/review case
External grade-sync job
Audit event

Every grade-related record should include:

Tenant ID
Course and term
Student/enrollment scope
Authorized grader
Rubric/version
Source calculation
Status: draft / graded / moderated / released / changed / synced
Timestamp
Reason for override/change
Audit correlation ID
Retention classification

What to build first

Phase 1: Work completion and study value

- Course Home
- Syllabus review/approval workflow
- Assignment planner and milestones
- Task/calendar integration
- Draft workspace with recovery
- Study Studio with source-grounded learning aids
- Office-hours, tutor, library, and writing-center handoffs
- Accessibility preferences and document checks

Phase 2: Native assessment foundation

- Assignment submission
- Rubrics
- Manual grading and feedback
- Basic gradebook
- Student grade view with estimate labels
- Grade-release controls
- Submission receipts and version history
- Accommodation-aware timing and availability
- QTI import/export foundation

Phase 3: Full instructor workflow

- Question banks
- Assessment builder
- Item pools and randomization
- Multiple assessment types
- Anonymous grading
- TA/calibration/moderation
- Group work and peer review
- Late-policy engine
- Grade-change and appeal workflow
- LTI AGS and SIS/LMS controlled sync

Phase 4: Advanced learning value

- Spaced repetition
- Adaptive practice, with transparent logic
- Skills/outcomes mapping
- Item analytics
- Portfolio/project assessment
- Credential evidence
- AI feedback assistance with human approval
- Course-level AI policy configuration

Phase 5: High-risk and specialized capabilities

Proctoring integrations
Secure browser
Coding sandboxes
Adaptive testing
Advanced psychometrics
Oral/video assessment
External credential issuance

Only launch these after you have dedicated accessibility testing, assessment-security controls, privacy/legal review, support capacity, and clear institutional operating models.

Release criteria for grading

Before Semester is used for consequential grading, require:

Named course/institution owner and authorized grader roles.
Rubric/version and assignment settings locked or versioned.
Student preview and accessible alternative path verified.
Accommodation rules applied and tested.
Submission autosave, receipt, recovery, and audit trail tested.
Grade calculations independently tested, including edge cases.
Grade visibility/release behavior tested from the student view.
Override and change workflows require reason and are audited.
No AI-only final grades or automated misconduct decisions.
Appeal/review process configured and visible to students.
LTI/SIS grade sync, if enabled, is previewable, permissioned, retry-safe, reconcilable, and reversible where the destination permits.
Backup, export, retention, and incident runbooks are tested.

The strongest learning platform is not the one with the most quizzes. It is one where students understand what they are learning, can complete work accessibly, receive useful feedback, recover from mistakes, and trust that every grade has a clear, reviewable human and system trail.



1. <https://www.1edtech.org/standards/lti>
2. https://standards.1edtech.org/lti/guides/implementation_guide/implementation-guide
3. <https://nfb.org/programs-services/center-excellence-nonvisual-access/higher-education-accessibility-online-resource>
4. <https://www.ahead.ie/top-10-tips-online-assessment>
5. <https://trefnycenter.mines.edu/resources/accessible-assessments/>
6. <https://ed.link/community/what-is-the-meaning-of-lti-integration/>
7. <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf>
8. <https://www.nist.gov/itl/ai-risk-management-framework>
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